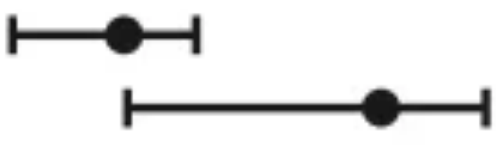


➔ **Magnitude Channels: Ordered Attributes**

Position on common scale



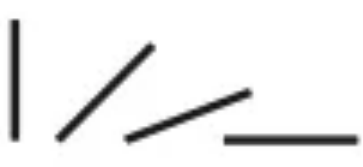
Position on unaligned scale



Length (1D size)



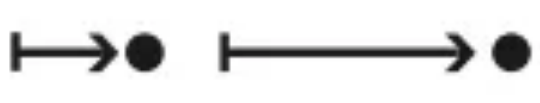
Tilt/angle



Area (2D size)



Depth (3D position)



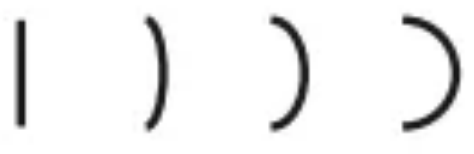
Color luminance



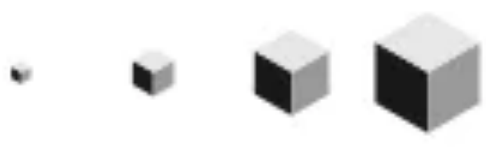
Color saturation



Curvature



Volume (3D size)



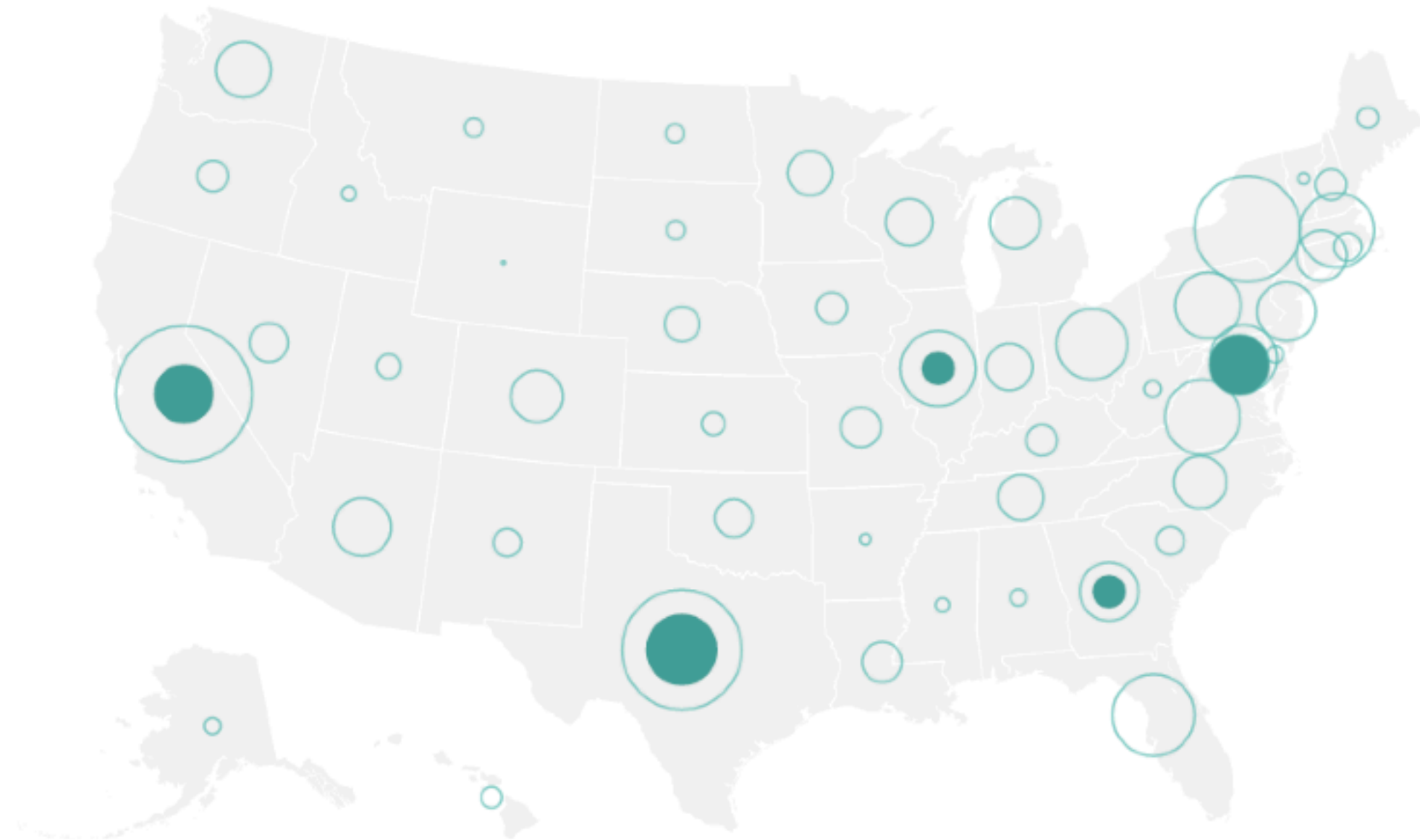
Same Same

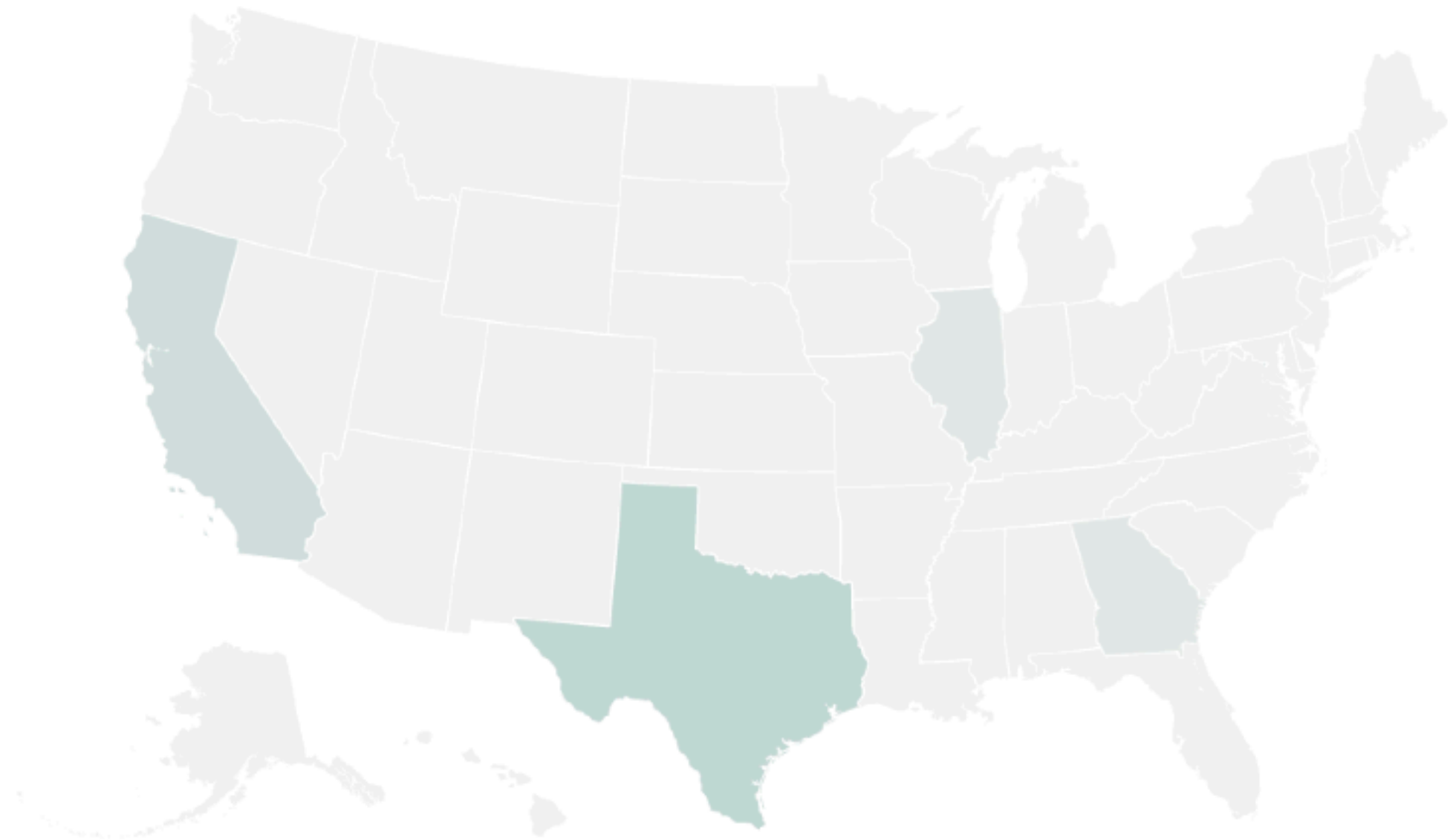
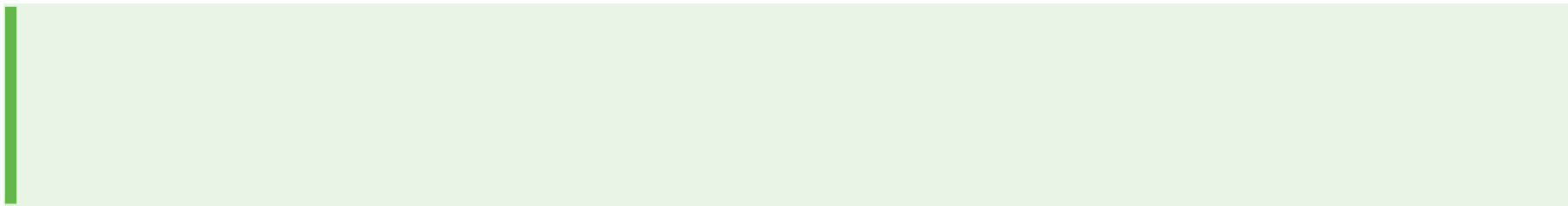
▲ Most
Effectiveness
Least ▼

2

3

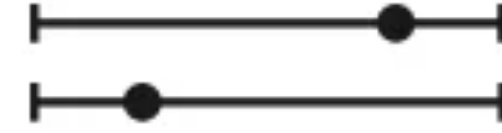
Tamara Munzner, Visualization Analysis and Design (2014)



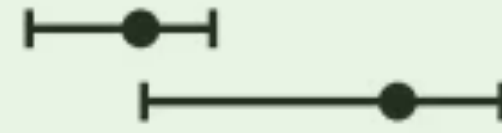


➔ Magnitude Channels: Ordered Attributes

Position on common scale



Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



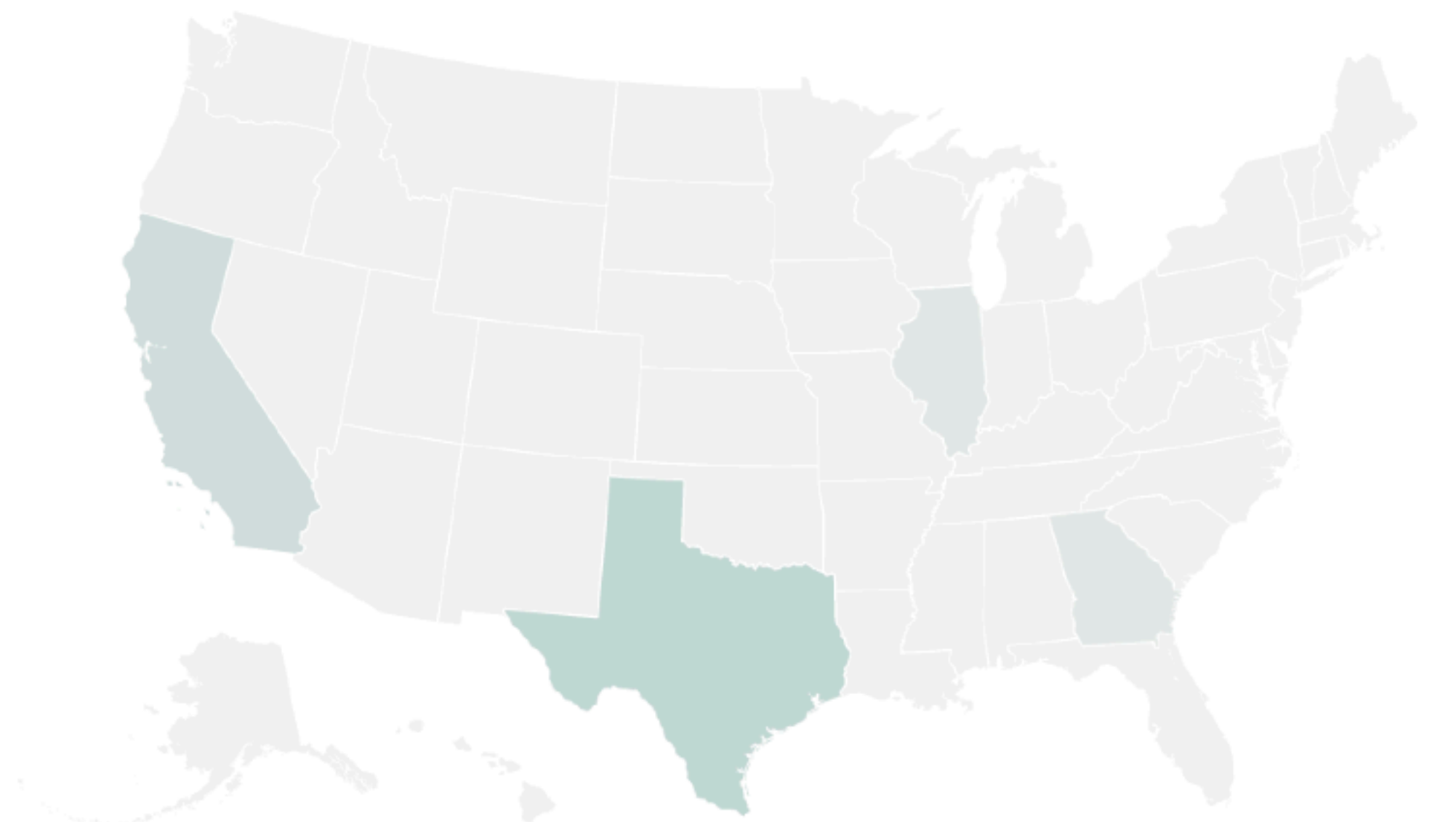
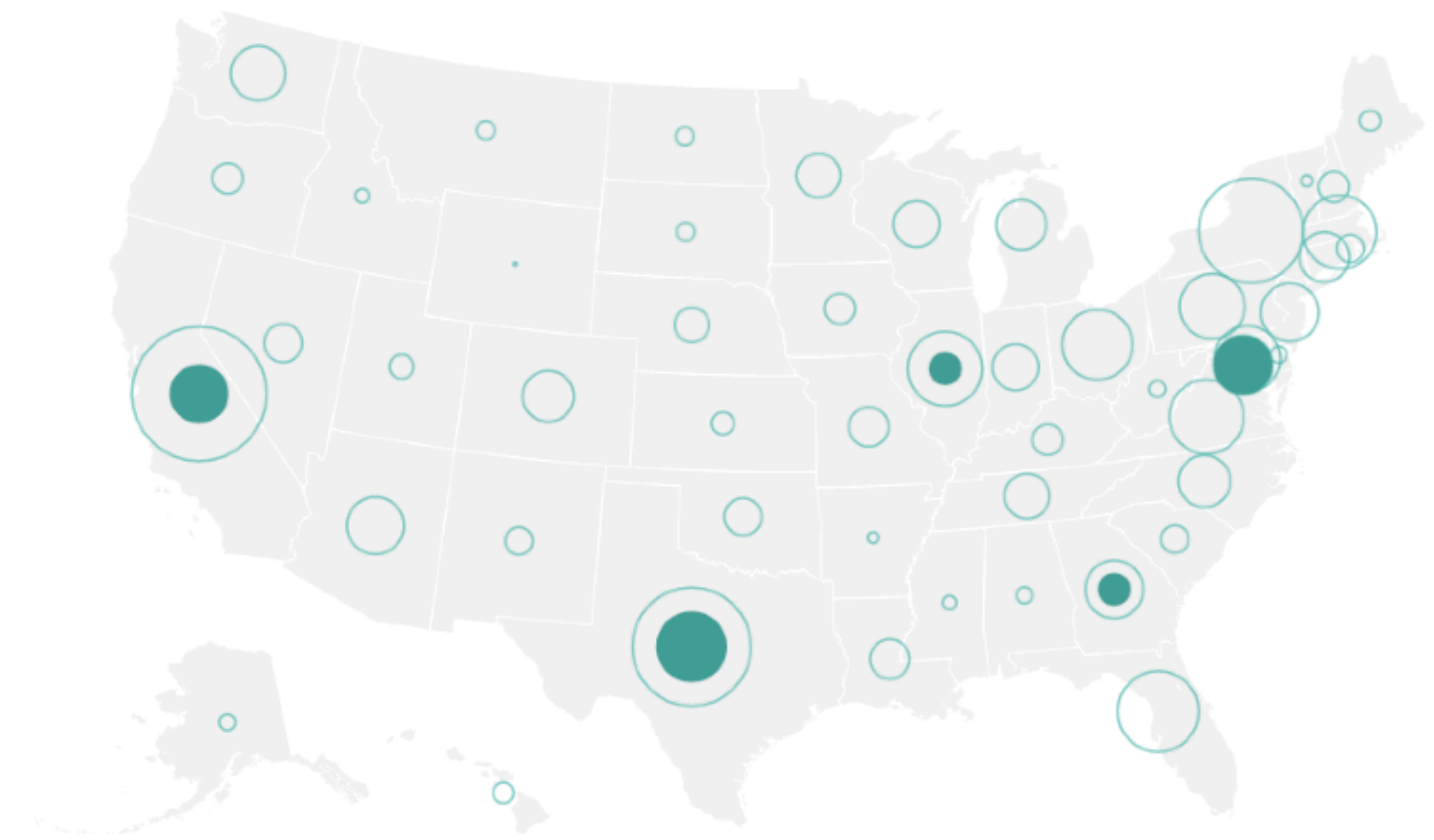
Volume (3D size)



Most

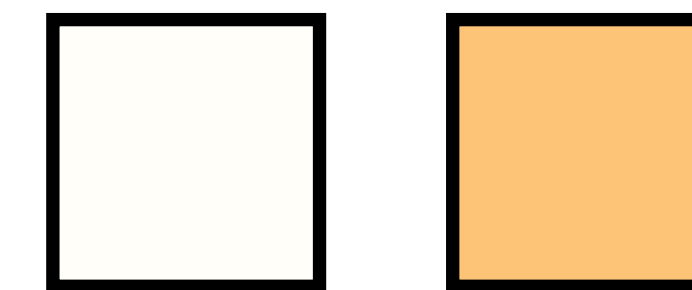
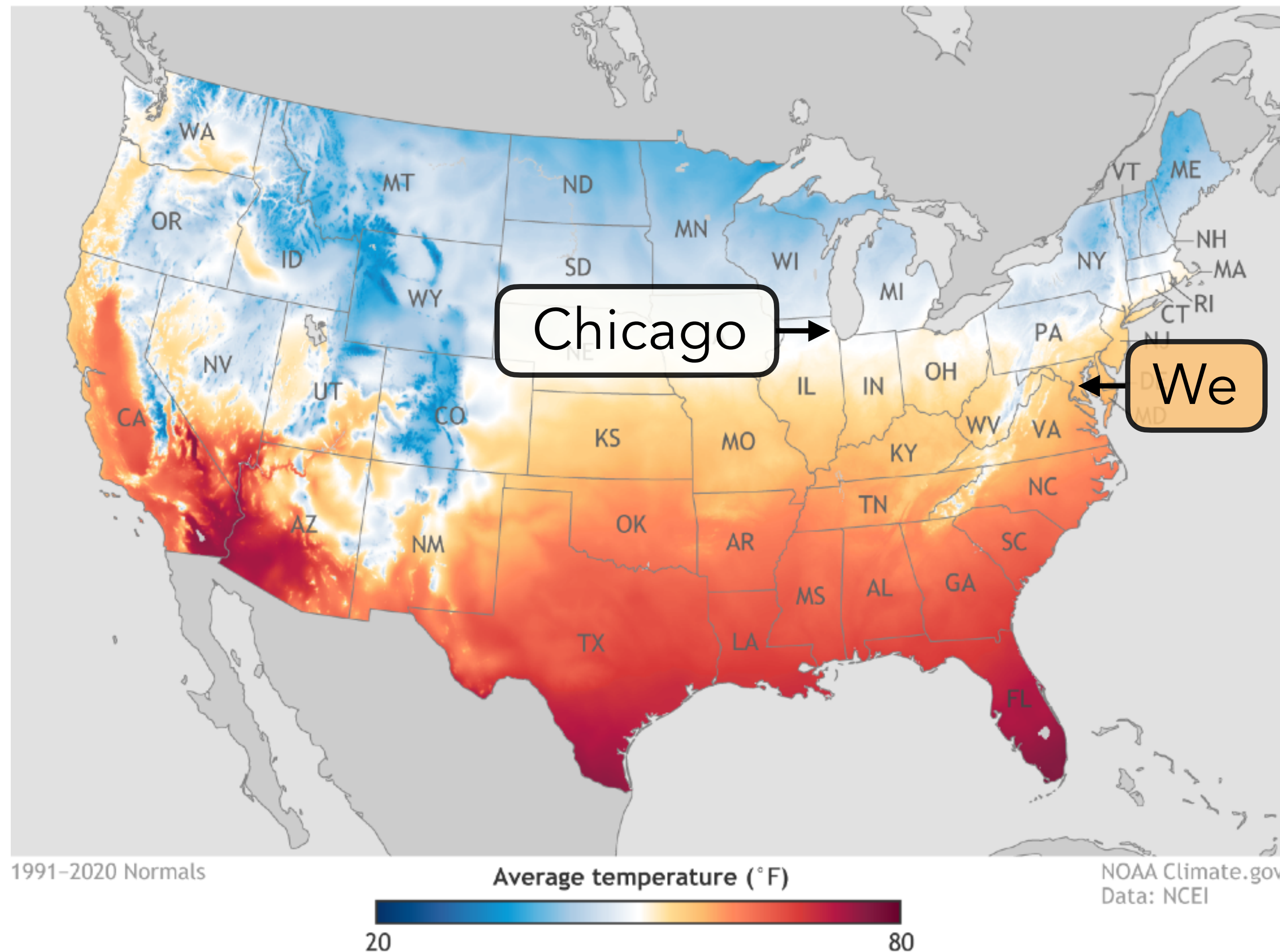
Effectiveness

Least



How many degrees warmer is College Park than Chicago?

U.S. annual average temperature and precipitation (1991–2020)



Not impossible

But difficult & prone-to-error for this task